

Ch 10: The Human Eye and the Colourful World

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SECTION A - INTRODUCTION TO THE HUMAN EYE AND ITS PARTS

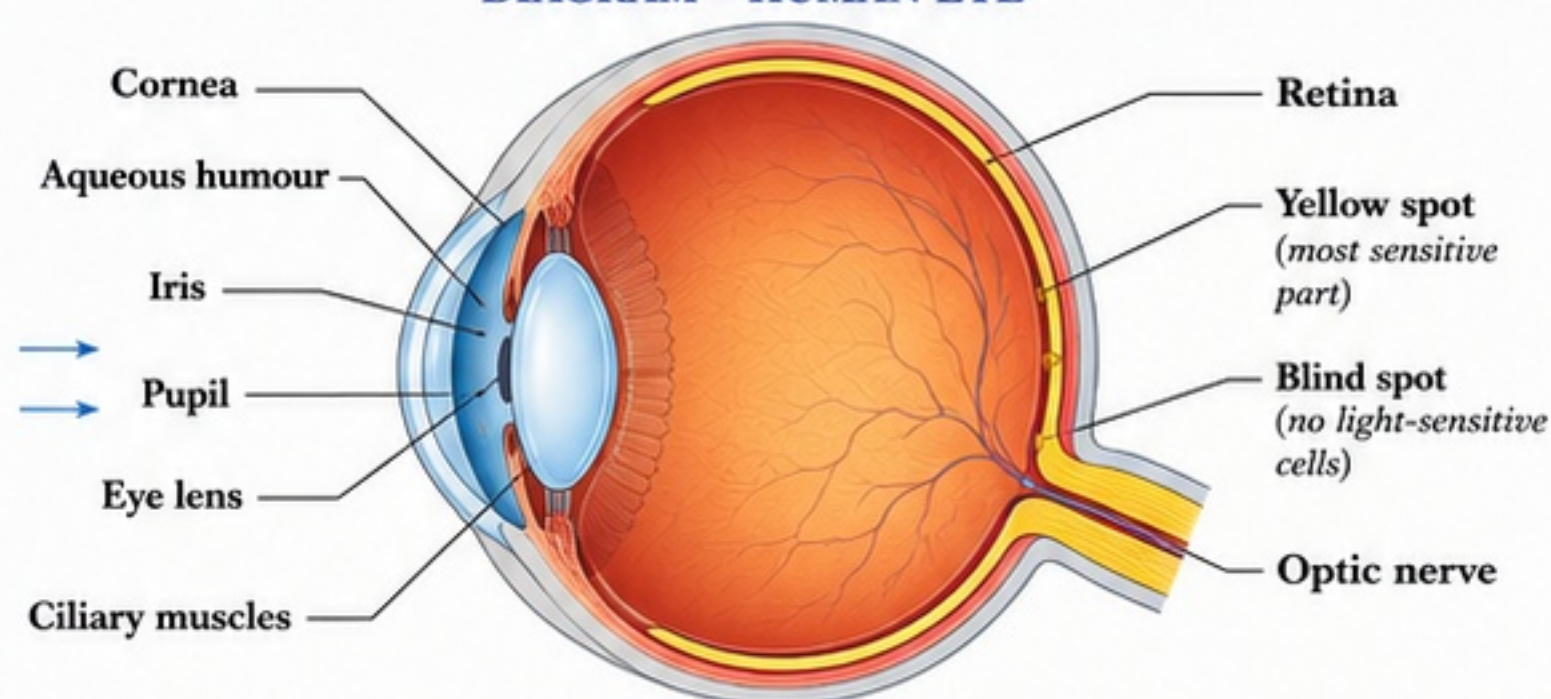
1. Why do we see?

- The human eye is the natural optical instrument of our body. Light reflected from objects enters the eye, gets focused and forms an image on the retina. The retina converts this light information into electrical signals and sends them through the optic nerve to the brain, where the image is interpreted.

2. Main functions of the human eye

- Vision** – helps us see objects and our surroundings.
- Brightness control** – adjusts the amount of light entering.
- Focusing** – adjusts to see near and far objects clearly.
- Colour recognition** – helps us distinguish different colours.

DIAGRAM – HUMAN EYE



One-line function of each major part

Cornea – Protects the eye and provides most refraction.
Aqueous humour – Nourishes the front parts and maintains shape.
Iris – Controls the size of the pupil and the amount of light.
Pupil – Opening through which light enters the eye.
Eye lens – Focuses light on the retina.

Ciliary muscles – Change the shape of the lens for focusing.
Vitreous humour – Maintains the spherical shape of the eye.
Retina – Contains light-sensitive cells and forms the image.
Yellow spot – Part of retina with the best vision.
Blind spot – Point where optic nerve leaves; no image forms.
Optic nerve – Carries signals from retina to the brain.

Parts of the Eye and Their Functions

Part	Function
Cornea	Transparent front covering; admits light and provides most of the refraction.
Iris	Coloured muscular diaphragm; controls the size of the pupil and thus the amount of light entering.
Pupil	Circular opening in the iris through which light enters the eye.
Eye lens	Transparent, flexible structure; changes shape to focus light on the retina.
Ciliary muscles	Muscles attached to the lens; contract or relax to change the curvature of the lens (accommodation).
Retina	Light-sensitive inner lining; contains rods and cones that form the image.
Optic nerve	Bundle of nerve fibres; carries electrical impulses from the retina to the brain.
Aqueous humour	Clear fluid between cornea and lens; nourishes and maintains pressure of the eye.
Vitreous humour	Jelly-like fluid behind the lens; maintains shape of the eyeball and presses retina gently.
Blind spot	Point on the retina where optic nerve leaves; no rods or cones are present.



IMPORTANT TERMS

- Pupil** – The opening in the centre of the iris that allows light to enter.
- Retina** – The light-sensitive layer at the back of the eye where the image is formed.
- Accommodation** – The ability of the eye lens to change its shape to see near or far objects clearly.
- Blind spot** – The point on the retina where the optic nerve leaves; no image is formed here.
- Persistence of vision** – The property of the retina by which the image remains for a fraction of a second even after the light stimulus disappears, allowing continuous vision.

5. Retina and image formation

- The eye lens forms a real, inverted and diminished image of an object on the retina. The retina detects this image and converts it into nerve impulses. The brain then interprets these signals and we perceive the image to be erect, of the correct size and in its true colour.



QUICK RECALL

- Cornea admits most refraction.
- Iris controls the amount of light.
- Pupil is the opening for light.
- Retina contains light-sensitive cells.
- Optic nerve carries impulses to the brain.



KEY IDEA

The human eye works like a camera—light enters through a small opening, is focused by a lens, and an image is formed on a light-sensitive screen (retina). But unlike a camera, the eye is a living, self-adjusting system that protects itself, supplies nourishment and sends the information to the brain for meaningful vision.



EXAM TIPS

- Draw the diagram of the eye neatly and label all parts correctly.
- Remember: The image formed on the retina is real, inverted and diminished.
- Write one clear function for each part of the eye.
- Use arrows to show the path of light when drawing diagrams.



CENTUM FOCUS

- Understand how each part works together for perfect vision.
- Practise neat, accurate diagrams.
- Relate structure with function while studying.
- Revise key terms regularly.
- Attempt diagram-based and application questions in exams.

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SECTION B - WORKING OF THE EYE, ACCOMMODATION AND VISION BASICS

1. How the eye forms an image

- Light from an object enters the eye through the cornea. It passes through the aqueous humour, the pupil, and the eye lens. Finally, it goes through the vitreous humour and forms a real, inverted and diminished image on the retina.

2. Role of iris and pupil in controlling light

- The iris controls the size of the pupil and thus regulates the amount of light entering the eye.

Condition	Pupil Size	Amount of Light Entering
Bright light	Constricts (becomes small)	Less
Dim light	Dilates (becomes large)	More

3. Accommodation

- Accommodation is the ability of the eye lens to adjust its curvature so that images of objects at different distances are formed clearly on the retina.
- Achieved by the action of **ciliary muscles**, which change the shape (thickness) of the eye lens.

4. Near point and far point of a normal eye

- Far point** – The farthest point an eye can see clearly without strain is at infinity.
- Near point** – The nearest point an eye can see clearly is 25 cm from the eye.
- Range of vision** – From the near point (25 cm) to infinity.

5. Power of accommodation

- The power of accommodation is the ability of the eye to change focal length for clear vision of objects at different distances.
- It is maximum for near objects and minimum for distant objects.
- It decreases with age because the eye lens becomes less elastic and the ciliary muscles become weaker.

6. Persistence of vision

- Persistence of vision is the property of the retina by which an image remains for a fraction of a second even after the stimulus is removed.
- It enables us to see continuous motion in movies, TV, animation and other moving images.

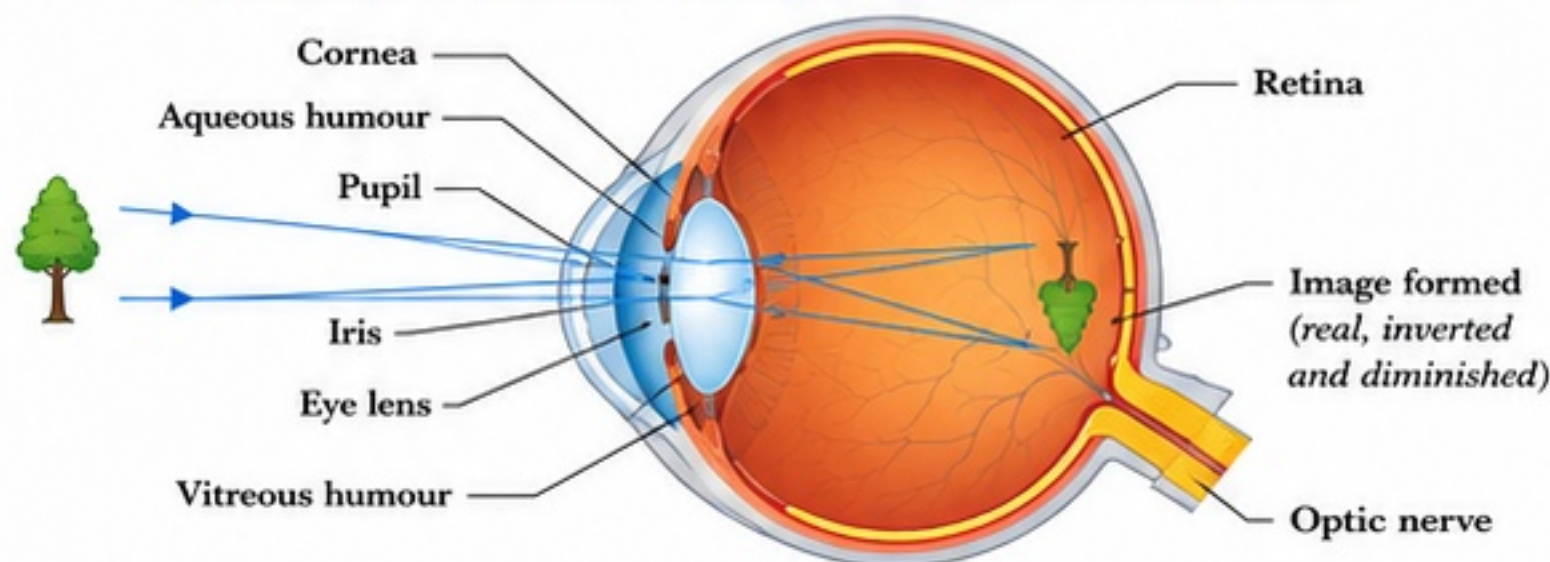
7. Cataract

- A cataract is the clouding of the eye lens due to aging or other reasons, causing blurred vision.
- It can be corrected by a simple surgery in which the cloudy lens is removed and an artificial intraocular lens is implanted.

8. Rods and cones in the retina

- Rods** – Sensitive to dim light; help in night vision; do not detect colours.
- Cones** – Work in bright light; detect colours; help in detailed and sharp vision.

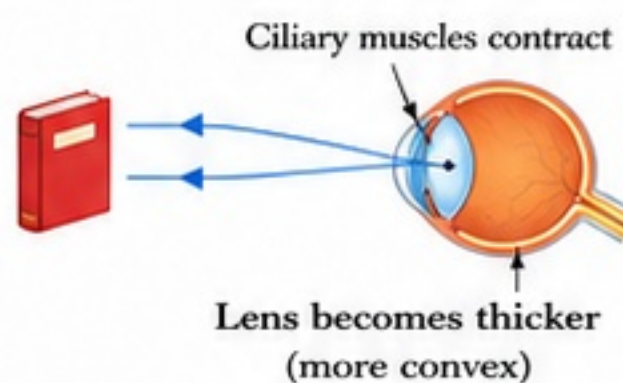
DIAGRAM – IMAGE FORMATION IN THE NORMAL EYE



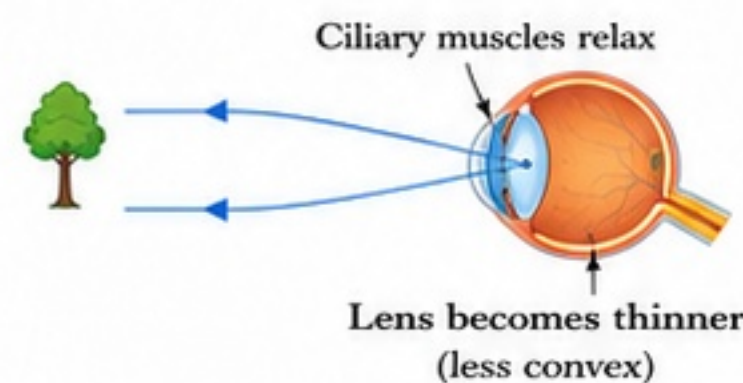
Light enters through the cornea → passes through aqueous humour → pupil → eye lens → vitreous humour and forms a real, inverted, diminished image on the retina.

ACCOMMODATION – FOCUSING FOR DIFFERENT DISTANCES

(a) Near object (at 25 cm)



(b) Distant object (at infinity)



NEAR OBJECT VISION VS DISTANT OBJECT VISION

Feature	Near Object Vision (≈25 cm)	Distant Object Vision (at infinity)
Ciliary muscles	Contract	Relax
Eye lens curvature	Increases (becomes thicker)	Decreases (becomes thinner)
Focal length of lens	Decreases	Increases
Power of lens	Increases	Decreases
Example	Reading a book	Looking at distant objects

IMPORTANT TERMS

- Accommodation** – Adjustment of the eye lens curvature to focus on objects at different distances.
- Far point** – The farthest point seen clearly without strain (at infinity for normal eye).
- Near point** – The nearest point seen clearly (25 cm for normal eye).
- Persistence of vision** – The ability of the retina to retain an image for a brief time even after the stimulus is removed.
- Cataract** – Clouding of the eye lens leading to blurred vision.
- Rods** – Light-sensitive cells for night vision (no colour).
- Cones** – Light-sensitive cells for colour vision and clear detail in bright light.



QUICK RECALL

- Image forms on retina: real, inverted and diminished.
- Iris controls pupil size.
- Accommodation changes lens shape.
- Near point = 25 cm; Far point = infinity.
- Cones for colour, rods for dim light.



KEY IDEA

- The eye adjusts the lens continuously for clear vision.
- Pupil size controls the amount of light.
- Range of normal vision is from 25 cm to infinity.
- Persistence of vision makes moving images possible.



EXAM TIPS

- Always draw neat diagrams for image formation and accommodation.
- Remember: Near object → lens thicker; Distant object → lens thinner.
- Learn the near point (25 cm) and far point (infinity) exactly.



CENTUM FOCUS

- Use tables to compare conditions clearly.
- Relate every term to its real-life example.
- Practise labelled diagrams regularly.
- Attempt long-answer questions on accommodation and vision.

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SECTION C - DEFECTS OF VISION OVERVIEW AND MYOPIA

1. Meaning of defect of vision

- A defect of vision is a condition in which the eye cannot form the image of objects clearly on the retina, either because of the shape of the eye or due to change in the focusing ability of the eye lens.

2. Why defects occur

- Change in eyeball length** – Eyeball may be longer or shorter than normal.
- Change in curvature/power of eye lens** – Lens may become too curved (more powerful) or too flat (less powerful).
- Age-related change** – With age, the eye lens loses elasticity and the ciliary muscles weaken, affecting focus.

3. Overview: Common defects in this chapter

We study three main defects:

- Myopia (Short-sightedness)
- Hypermetropia (Long-sightedness)
- Presbyopia (Age-related defect)

MYOPIA (SHORT-SIGHTEDNESS)

4. What is myopia?

- Myopia (short-sightedness) is a defect in which a person can see nearby objects clearly but cannot see distant objects clearly.
- The far point of a myopic eye comes closer to the eye (i.e. not at infinity).
- The image of a distant object is formed in front of the retina instead of on it.

5. Causes of myopia

- Elongation of the eyeball** – The eyeball becomes longer than normal.
- Excessive converging power of the eye lens** – The lens becomes too curved or ciliary muscles remain contracted, increasing the power of the lens.

(A) Myopic Eye – Image forms before retina

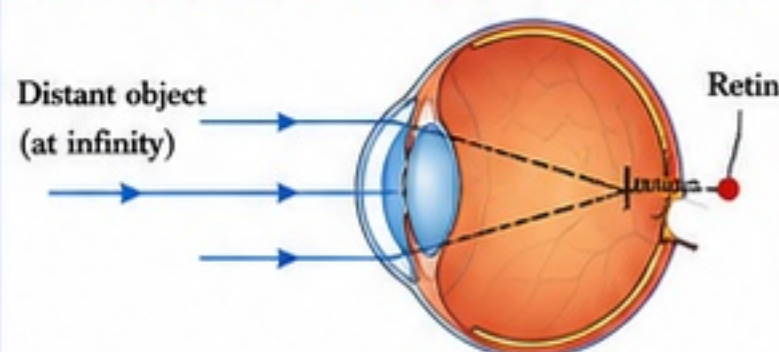
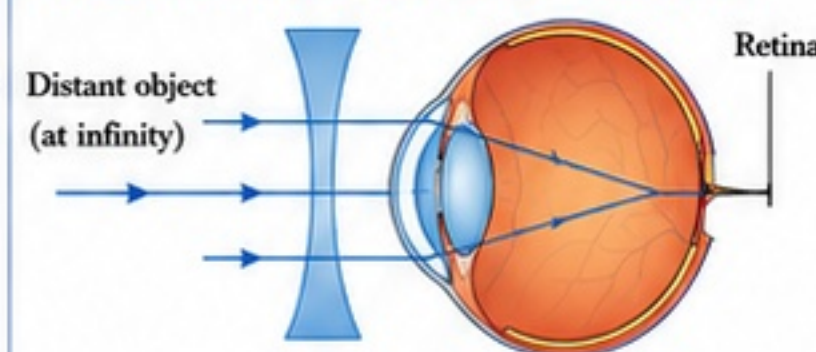


Image is formed in front of the retina → distant object appears blurred.

(B) Correction of Myopia using Concave Lens



Concave lens diverges the incoming rays so that the image is formed on the retina.

6. Normal eye vs Myopic eye

Feature	Normal Eye	Myopic Eye
Eyeball length	Normal	Longer than normal
Power of eye lens	Normal	Too high (excessive)
Far point	At infinity	At a finite distance closer than infinity
Image of distant object	Forms on the retina	Forms in front of the retina
Vision	Distant and near objects seen clearly	Near objects clear; distant objects blurred
Correction	Not required	Concave (diverging) lens

7. Correction of myopia

- Myopia is corrected using a concave lens (diverging lens).
- The concave lens diverges the incoming parallel rays from a distant object.
- This reduces the convergence by the eye lens and shifts the focus backward so that the image forms on the retina.
- Power of concave lens is chosen such that images of distant objects are focused on the retina.

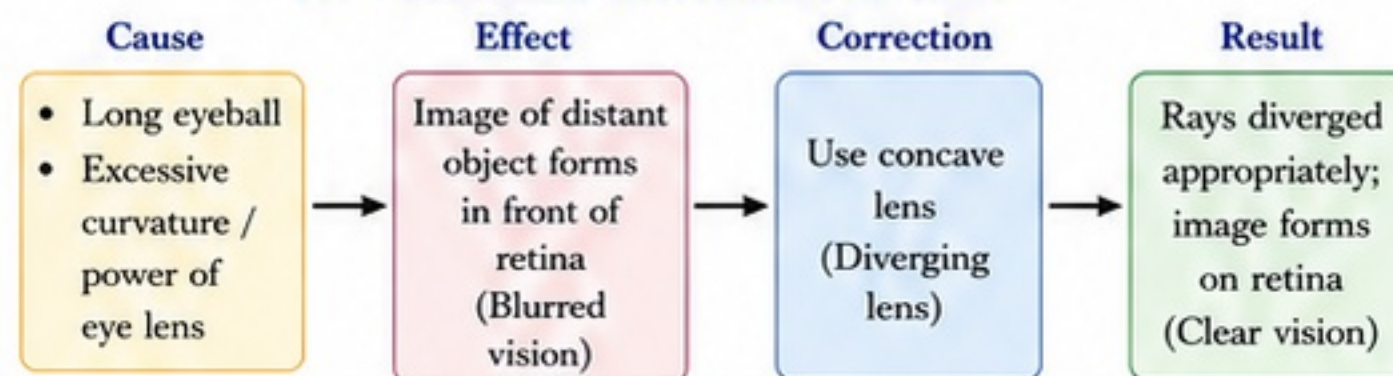
8. Numerical / Concept point

- Concave lenses have negative power.

9. Practical examples / Signs of myopia

- A myopic person sits close to the blackboard or television while watching.
- Difficulty in seeing road signs, distant buildings, the screen in a classroom.
- Often squints eyes to see distant objects.
- May experience eye strain or headaches after focusing on distant objects.

10. Cause-and-Correction Flowchart



QUICK RECALL

- Myopia = short-sightedness.
- Sees near objects clearly, distant objects not clearly.
- Image of distant object forms in front of retina.
- Corrected using concave lens.

KEY IDEA

Myopia occurs when the eye is too strong (high converging power) or the eyeball is too long. A concave lens 'pushes' the focus back so that the image of a distant object forms exactly on the retina, restoring clear vision.

EXAM TIPS

- Always draw rays parallel for distant objects.
- Mark where the image is formed (before retina in myopia).
- Note: Concave lens always has negative power.

CENTUM FOCUS

- Compare normal eye and myopic eye clearly.
- Learn cause → effect → correction → result flow.
- Practise ray diagrams neatly and label all parts.

SECTION D - HYPERMETROPIA, PRESBYOPIA AND THEIR CORRECTION

1. Hypermetropia (Long-sightedness)

- **Hypermetropia** is a common defect of vision in which a person cannot see **nearby** objects clearly, while distant objects may be seen clearly.
- The near point of the eye moves farther away than the normal near point of 25 cm.
- When we look at a near object, the image is formed **behind** the **retina**, so it cannot be seen clearly.

2. Causes of Hypermetropia

- Eyeball is too short from front to back.
- Insufficient converging power of the eye lens due to weak ciliary muscles or eye lens too flat.

3. Correction of Hypermetropia

- It is corrected by using a **converging (convex) lens** of suitable power.
- The convex lens converges the incident light rays and forms a virtual image of the near object at the near point of the eye (or within the range of accommodation).
- The eye lens then forms the real image of this virtual object on the retina, giving clear vision.

4. Presbyopia (Age-related Long-sightedness)

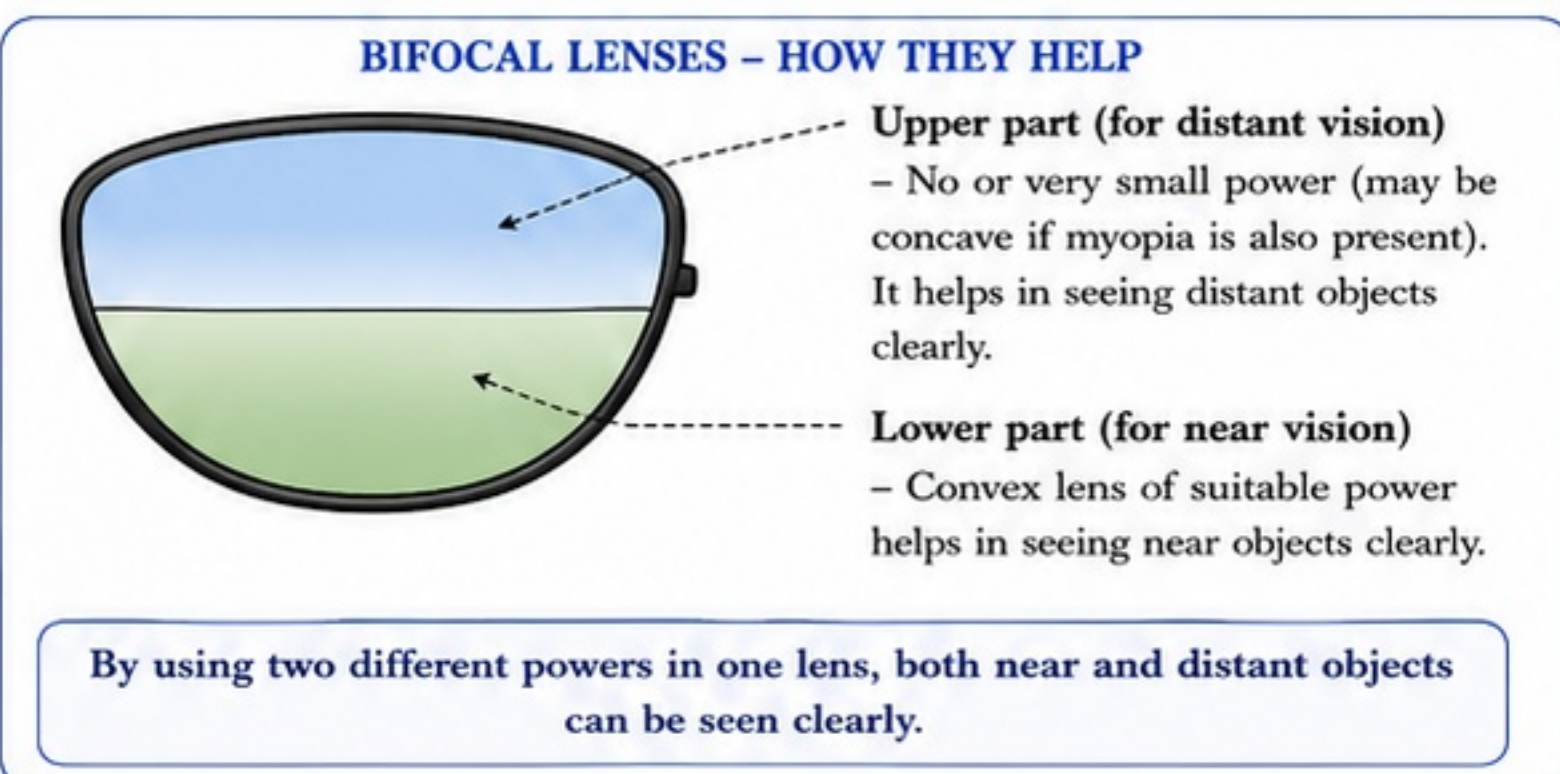
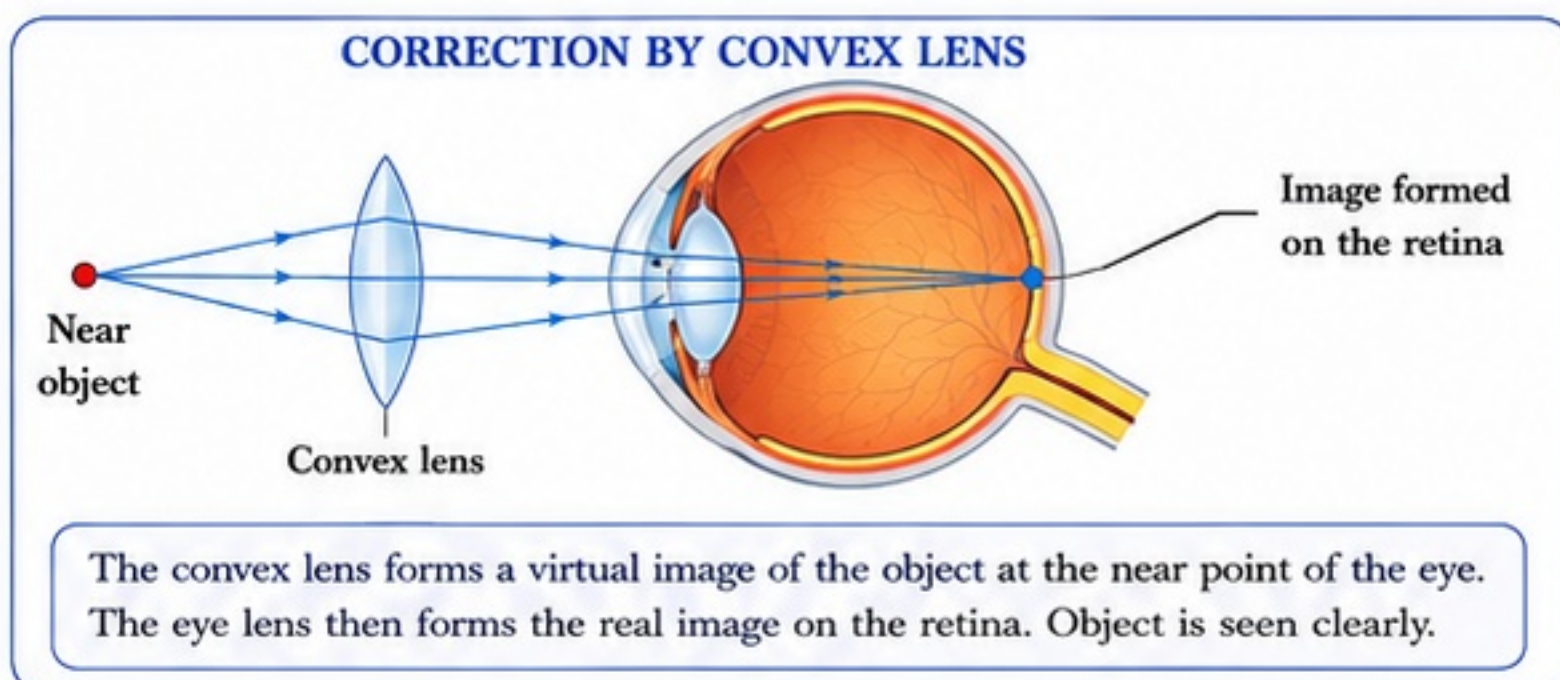
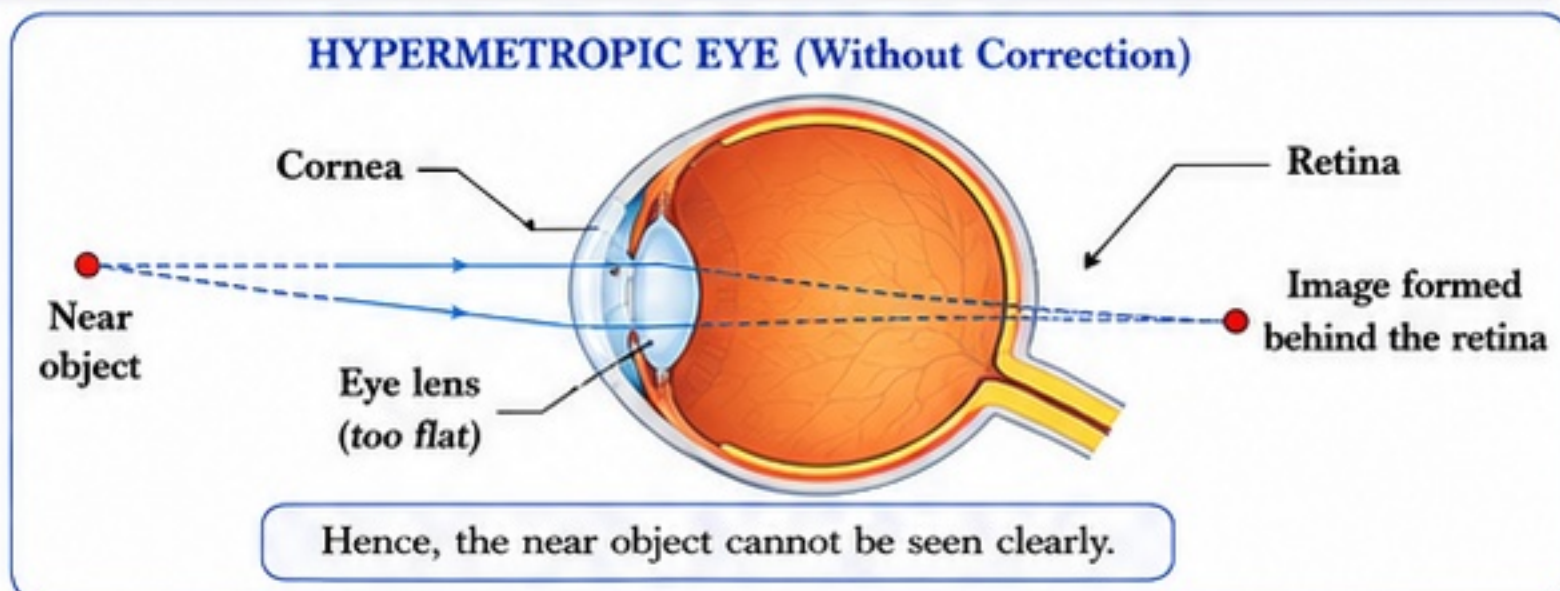
- Presbyopia is an age-related defect of vision.
- With advancing age, the ciliary muscles become weak and the eye lens loses its flexibility.
- The eye loses its power of accommodation, making it difficult to see nearby objects clearly.
- The near point shifts farther away from the eye.
- Presbyopia may occur along with myopia or hypermetropia.

5. Correction of Presbyopia

- It is corrected by using a convex lens for near vision.
- For persons who also have myopia or hypermetropia, **bifocal lenses** or **progressive addition lenses** are used.

6. Comparison: Myopia, Hypermetropia and Presbyopia

Feature	Myopia (Short-sightedness)	Hypermetropia (Long-sightedness)	Presbyopia (Age-related)
Objects seen clearly	Near objects	Distant objects	Distant objects (near objects not seen clearly)
Objects not seen clearly	Distant objects	Near objects	Near objects
Near point	Less than 25 cm (nearer to eye)	More than 25 cm (farther from eye)	Moves farther than 25 cm with age
Image of near object forms	In front of the retina	Behind the retina	Behind the retina (due to reduced accommodation)
Cause	Eyeball too long / eye lens too curved (or cornea too curved)	Eyeball too short / eye lens too flat (or weak ciliary muscles)	Weak ciliary muscles and loss of flexibility of eye lens with age
Correction	Concave (diverging) lens	Convex (converging) lens	Convex lens for near vision; bifocal or progressive lenses if combined defects are present



QUICK RECALL

- **Hypermetropia:** near objects not seen clearly; image behind retina; corrected by convex lens.
- **Presbyopia:** age-related loss of accommodation; near objects not seen clearly; corrected by convex or bifocal lenses.
- Near point in hypermetropia and presbyopia shifts beyond 25 cm.



KEY IDEA

- In hypermetropia, use a convex lens to bring the image forward onto the retina.
- In presbyopia, weak ciliary muscles reduce accommodation; convex lens helps for near vision.
- Bifocal lenses combine two powers: top for distant, bottom for near vision.



EXAM TIPS

- Always mention position of image (in front / behind retina).
- State the cause clearly (eyeball length or lens power).
- Mention the type of lens used and its role in correction.
- Draw neat, labelled ray diagrams for marks.



CENTUM FOCUS

- Differentiate defects by what is seen clearly and where the image forms.
- Learn bifocal lenses diagram and their use in combined defects.
- Practise diagrams and tabular comparisons regularly.

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SECTION E - REFRACTION THROUGH A PRISM, DISPERSION AND SPECTRUM

1. Recall of refraction

- Refraction is the bending of light when it passes from one medium to another due to a change in the speed of light.

2. Triangular prism

- A triangular prism is a transparent block of glass bounded by three rectangular faces and having a triangular cross-section.
- When light enters the first face (air to glass), it bends towards the normal. When it emerges from the second face (glass to air), it bends away from the normal.

3. Refraction through a prism and deviation

- A narrow beam of white light incident on one face of a prism emerges from the other face after refraction.
- The emergent beam is deviated from its original direction. The angle between the incident direction and emergent direction is called angle of deviation.

4. White light and its constituent colours

- White light is a mixture of seven colours. Each colour has a different wavelength and speed in a medium (like glass).

5. Dispersion of white light by a prism

- Dispersion is the splitting of white light into its constituent colours when it passes through a prism.
- It occurs because different colours refract by different amounts in the prism.

6. Spectrum and VIBGYOR

- The band of seven colours obtained on a screen after dispersion is called the spectrum.
- The order of colours in the spectrum is V I B G Y O R (from top to bottom).

7. Why different colours deviate by different amounts

- Different colours have different wavelengths and hence different speeds in glass.
- Violet travels slowest in glass → deviates most.
- Red travels fastest in glass → deviates least.

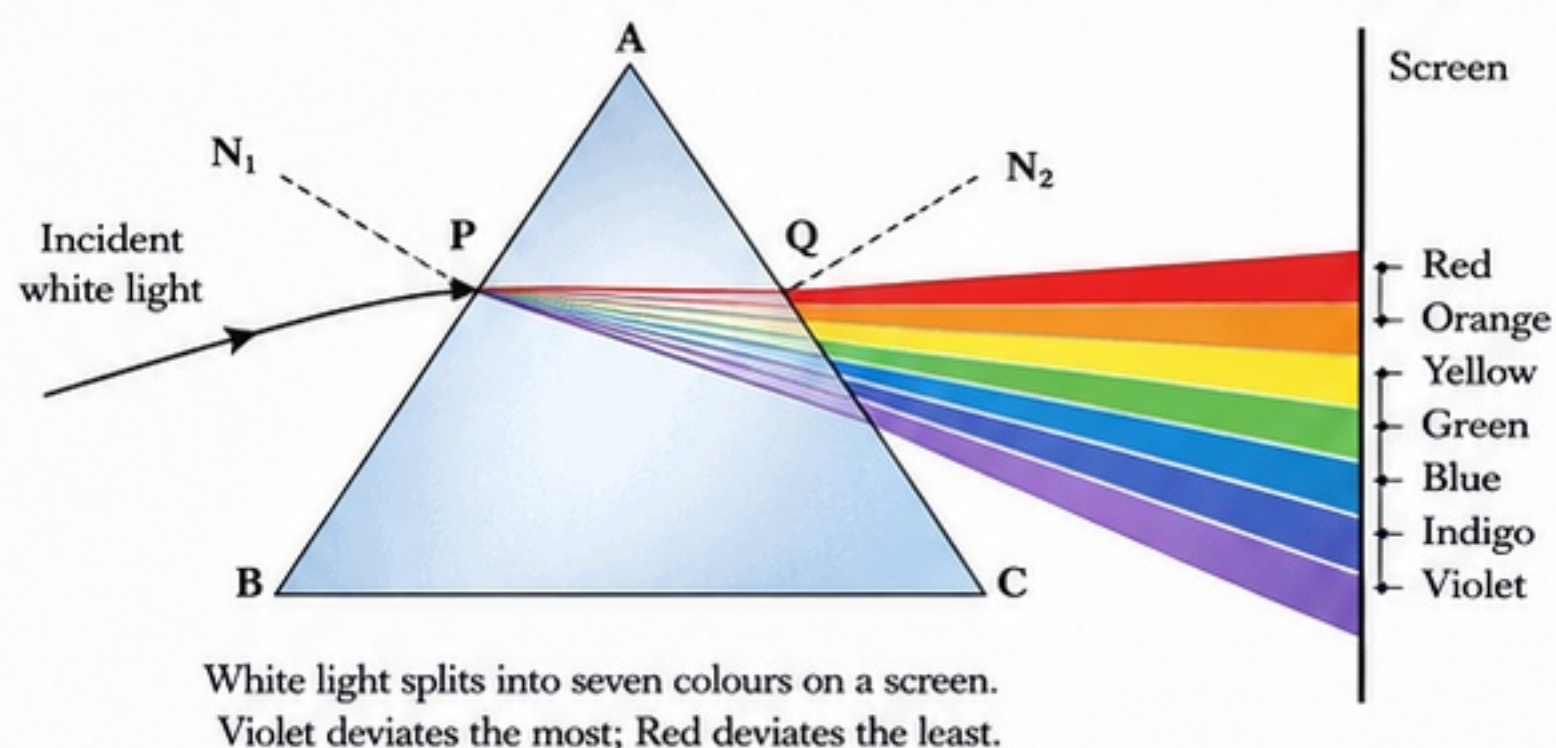
8. Recombination of colours to form white light

- If the dispersed colours are passed through an inverted prism, they recombine to form a single beam of white light.
- This shows that white light is made of these seven colours.

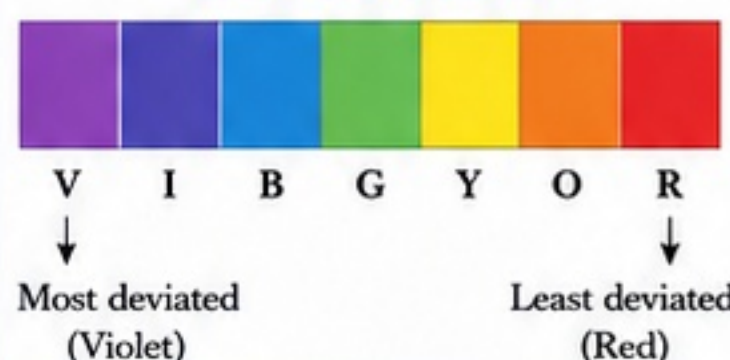
9. Real-life significance and examples

- Basis of rainbows (natural dispersion in raindrops).
- Used in spectroscopes, colour TV/monitors, photography and in scientific analysis of light from stars.
- Helps in understanding colours of sky during sunrise/sunset.

REFRACTION THROUGH A PRISM – DISPERSION OF WHITE LIGHT



VIBGYOR SPECTRUM



RECOMBINATION OF COLOURS

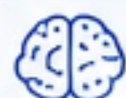


REFRACTION vs DISPERSION

Aspect	Refraction	Dispersion
Definition	Bending of light when it passes from one medium to another due to change in speed.	Splitting of white light into its constituent colours due to different refraction of colours.
Cause	Change in speed of light between two mediums.	Different colours travel at different speeds in the same medium (like glass).
Result	Direction of light changes; colour remains the same.	White light splits into seven colours.
Example	Bending of a pencil in water.	Formation of rainbow or spectrum by a prism.

COLOURS OF THE SPECTRUM AND THEIR RELATIVE DEVIATION

Order	Colour	Wavelength (approx.) (nm)	Relative Deviation in Prism	
1	Violet (V)	380 – 450	Most deviation	Most ↓ Least
2	Indigo (I)	450 – 475	More	
3	Blue (B)	475 – 495	More	
4	Green (G)	495 – 570	Moderate	
5	Yellow (Y)	570 – 590	Less	
6	Orange (O)	590 – 620	Less	
7	Red (R)	620 – 750	Least deviation	



QUICK RECALL

- Refraction – bending of light due to change in speed.
- Dispersion – splitting of white light into seven colours.
- Spectrum order – V I B G Y O R.
- Violet deviates most, Red least.
- Colours can recombine to form white light again.



KEY IDEA

White light is a mixture of seven colours. Different colours have different wavelengths and speeds in glass, so they refract (deviate) by different amounts. A prism separating colours and an inverted prism joining them back are strong evidences that white light is not a single colour but a combination.



EXAM TIPS

- Always remember the order V I B G Y O R.
- Draw a neat prism diagram with correct bending and spectrum.
- State that violet deviates most and red least.
- For recombination, show inverted prism with colours merging into white light.



CENTUM FOCUS

- Practise prism diagrams until angles and labels are perfect.
- Learn the difference between refraction and dispersion.
- Use tables in answers for colours and deviation.
- Relate dispersion to rainbows and real-life applications.

SECTION F - ATMOSPHERIC REFRACTION AND ITS PHENOMENA

1. What is atmospheric refraction?

- Atmospheric refraction is the bending of light from celestial objects as it passes through Earth's atmosphere, which has layers of air with gradually changing density and hence varying refractive index.

2. Why atmospheric layers have different refractive indices

- Air is not uniform. Near the Earth's surface it is denser (higher pressure) and contains more water vapour and dust.
- With increase in altitude, pressure, density and temperature change, making air rarer.
- Since refractive index (n) increases with density, lower layers have higher n and upper layers have lower n .
- Thus, light travelling upward from a distant object goes from higher n to lower n gradually and bends towards the normal at each layer, following a curved path.

3. Twinkling of stars

- Starlight passes through many turbulent layers of air with constantly fluctuating density. The amount and direction of refraction keep changing rapidly.
- This causes quick changes in the apparent brightness and position of stars – they appear to twinkle.
- Planets subtend a larger apparent size and their light from different parts averages out the fluctuations, so they generally do not twinkle.

4. Apparent position of stars due to atmospheric refraction

- Because of refraction, a star appears higher than its actual position.
- The effect is maximum when the star is near the horizon and zero when the star is at the zenith.
- Hence, stars are seen for a longer time than they actually are above the horizon.

5. Advanced sunrise and delayed sunset

- When the Sun is just below the horizon, its light from the upper rays enters Earth's atmosphere.
- These rays are continuously refracted towards the normal at each layer and finally reach the observer.
- Therefore, the Sun is visible about 2 minutes before the actual sunrise and about 2 minutes after the actual sunset.
- During this interval the Sun is below the horizon, but appears above it due to atmospheric refraction.

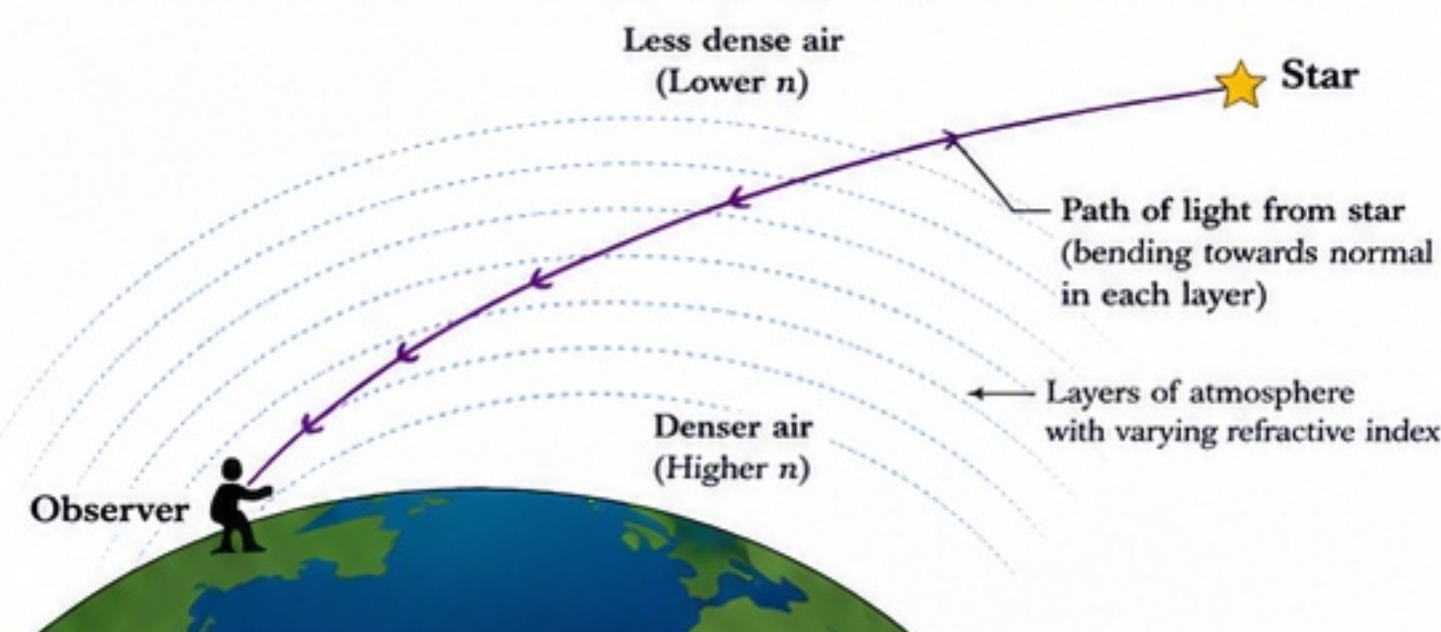
6. Apparent flattening of the Sun at sunrise and sunset

- The lower limb of the Sun is affected by greater refraction than the upper limb at the horizon.
- Hence, the lower limb is lifted more, making the Sun appear as an oval (flattened at the bottom) during sunrise and sunset.

7. Link: changing optical density with altitude

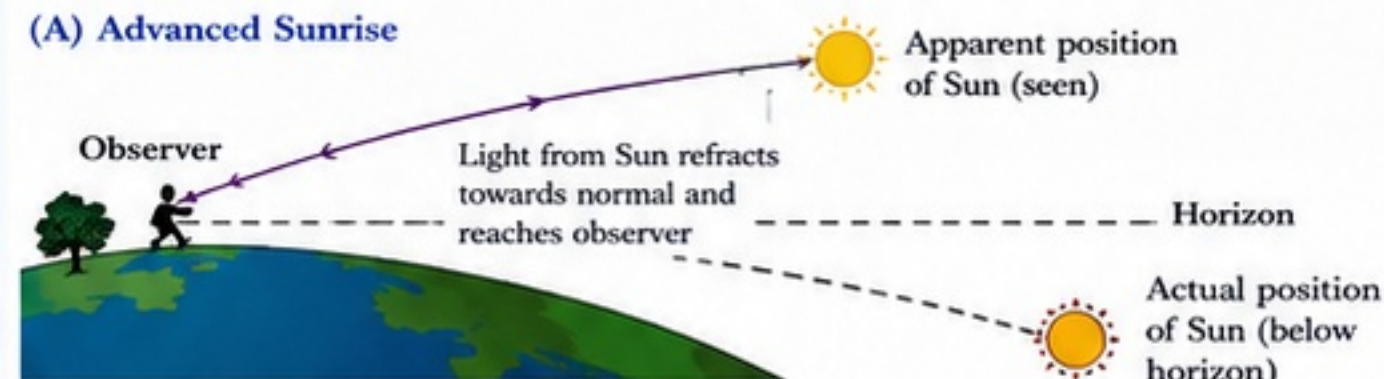
- Optical density ($\propto$ number of molecules per unit volume) is maximum near the surface and decreases with altitude.
- Lower layers (high optical density) have higher refractive index; upper layers (low optical density) have lower refractive index.
- This continuous change in optical density with altitude is the fundamental cause of atmospheric refraction and all its phenomena.

REFRACTION OF STARLIGHT BY EARTH'S ATMOSPHERE

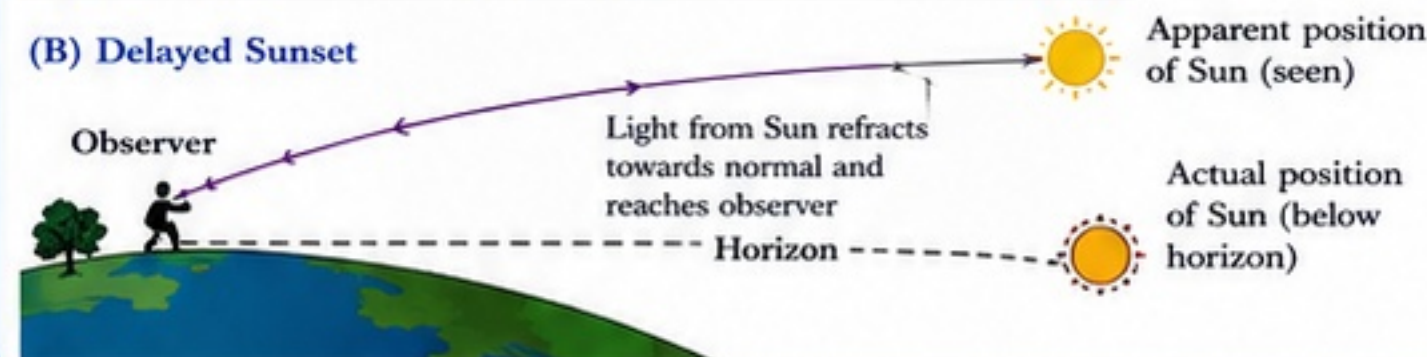


ADVANCED SUNRISE AND DELAYED SUNSET

(A) Advanced Sunrise



(B) Delayed Sunset



TWINKLING OF STARS: STARS vs PLANETS

Feature	Stars	Planets
Apparent size	Point-like sources	Have appreciable apparent size
Light path	From a point source	From many points across the disc
Atmospheric effect	Strong fluctuations in refraction	Fluctuations average out
Brightness	Varies rapidly	Nearly constant
Twinkling	Yes	Generally No
Reason	Light from a point is strongly scattered by turbulent layers	Light from extended disc averages out variations

PHENOMENA OF ATMOSPHERIC REFRACTION

Phenomenon	Explanation
Stars seen higher above horizon	Refraction bends starlight towards normal, making stars appear higher than their actual position.
Stars visible for longer duration	Stars appear above horizon even when actually below it due to refraction.
Twinkling of stars	Rapid changes in refractive index of air cause fluctuations in brightness and position.
Planets do not twinkle	Larger apparent size of planets averages out the fluctuations.
Advanced sunrise	Sun appears about 2 minutes before actual sunrise.
Delayed sunset	Sun remains visible about 2 minutes after actual sunset.
Apparent flattening of Sun	Greater refraction of lower limb lifts it more than upper limb, so Sun appears flattened at horizon.



QUICK RECALL

- Atmospheric refraction bends light towards normal.
- Refractive index decreases with altitude.
- Stars twinkle; planets do not.
- Sun is seen ~2 min earlier in morning and ~2 min later in the evening.
- Sun looks flattened at sunrise and sunset.



KEY IDEA

The atmosphere acts like a natural prism with continuously changing refractive index. Light from celestial objects travels along a curved path. This explains why we see objects higher, for longer duration, earlier in morning and later in evening, and why stars twinkle.



EXAM TIPS

- Draw ray diagrams for sunrise, sunset and starlight refraction.
- Remember: refraction is maximum near horizon and zero at zenith.
- Value to remember: Sun is seen about 2 minutes earlier and 2 minutes later due to refraction.
- Mention change in density $\rightarrow$ change in refractive index.



CENTUM FOCUS

- Explain twinkling using turbulent layers and variable refraction.
- Add numeric value (2 minutes) for sunrise and sunset.
- Note the reason for flattened appearance of Sun.
- Connect concepts: density $\rightarrow$ refractive index $\rightarrow$ refraction $\rightarrow$ phenomena.

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SECTION G - SCATTERING OF LIGHT AND THE TYNDALL EFFECT

1. Meaning of scattering of light

- Scattering of light is the phenomenon in which light gets deflected in different directions when it strikes tiny particles of matter such as molecules of air, dust, smoke or colloidal particles. As a result, the path of light becomes visible and the direction of light may change.

2. Tyndall effect

- Definition** – The scattering of light by colloidal particles such that the path of the light becomes visible is called the Tyndall effect.
- Scattering by colloidal particles** – Colloidal particles are larger than molecules but smaller than the particles in suspension. They are capable of scattering light strongly.
- Examples in daily life** – The beam of light from a torch becomes visible in a dark, dusty room; sunlight entering a dense forest appears as bright shafts; headlights in fog are seen clearly.

3. Why the sky appears blue

- Sunlight is a mixture of many colours (VIBGYOR). When sunlight enters the Earth's atmosphere, it is scattered by air molecules in all directions. Shorter wavelengths such as blue scatter much more than longer wavelengths such as red. Hence, blue light from all directions enters our eyes, making the sky appear blue.

4. Why the Sun appears reddish at sunrise and sunset

- During sunrise and sunset, sunlight has to travel a much longer path through the atmosphere. Most of the shorter wavelengths (blue, violet) are scattered away. Only longer wavelengths (red, orange) are scattered less and reach our eyes. Hence, the Sun appears reddish.

5. Why danger signals are red

- Red light has the longest wavelength and scatters the least in the atmosphere. Therefore, it can be seen from a much larger distance. For this reason, danger signals, tail lights and stop signs are made red.

6. Scattering and colour of clouds, fog and mist

- Clouds, fog and mist contain tiny water droplets much larger than air molecules. These droplets scatter all wavelengths of visible light almost equally. The mixture of all colours in equal proportion appears white.

7. Relation between wavelength and scattering

- Scattering is inversely related to the fourth power of wavelength (Rayleigh's law). In simple terms:

Shorter wavelength → Scatters more | Longer wavelength → Scatters less

DIAGRAM 1 – WHY THE SKY IS BLUE

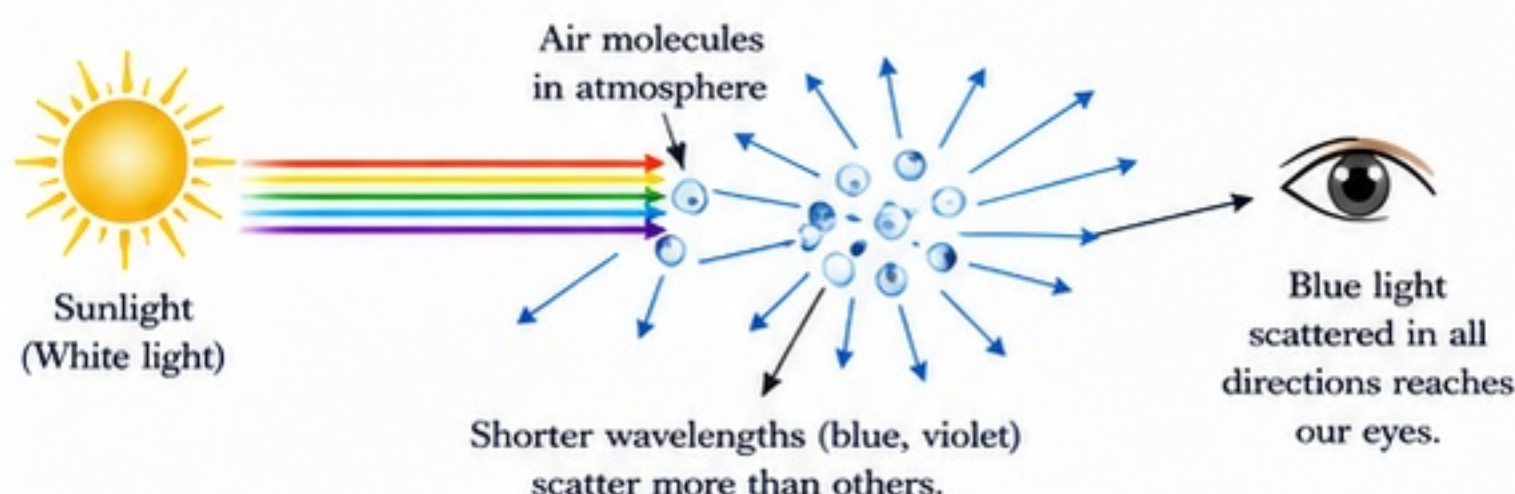


DIAGRAM 2 – PATH LENGTH OF SUNLIGHT

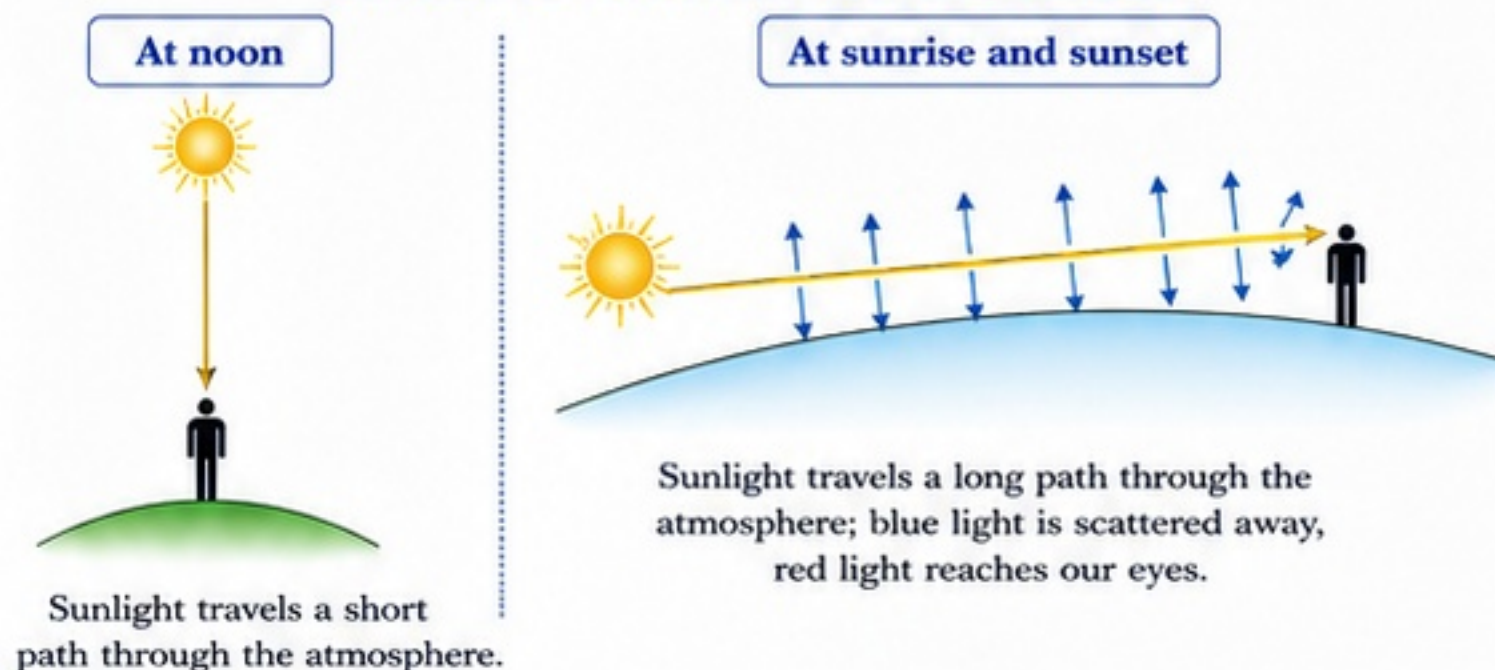


DIAGRAM 3 – THE TYNDALL EFFECT

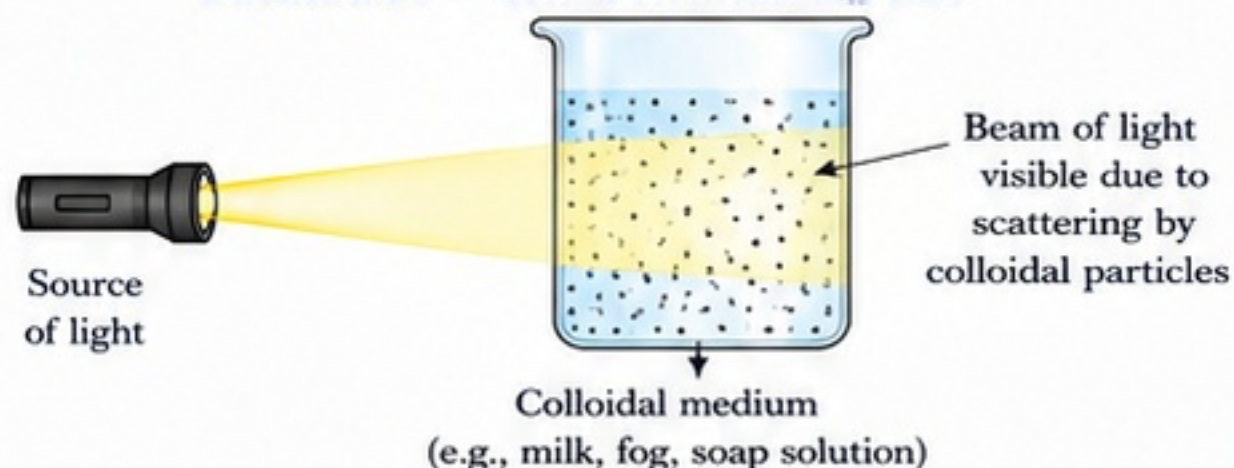


TABLE – SCATTERING IN EVERYDAY PHENOMENA

Phenomenon	Cause of Scattering	Explanation
Blue sky	Scattering by air molecules	Shorter wavelengths (blue, violet) scatter more in all directions and enter our eyes.
Reddish Sun at sunrise/sunset	Longer path through atmosphere	Blue light is scattered away; longer wavelengths (red, orange) scatter less and reach our eyes.
Red danger signals	Least scattering of red light	Red light has the longest wavelength; it scatters least and can be seen from a larger distance.



QUICK RECALL

- Scattering makes the path of light visible.
- Tyndall effect occurs in colloids.
- Blue sky – shorter wavelengths scatter more.
- Red Sun at sunrise/sunset – longer path, blue scattered away.
- Red signals – scatter least, seen far.
- Clouds/fog/mist look white.



KEY IDEA

Scattering depends on the size of the particles and the wavelength of light. Smaller particles (air molecules) scatter shorter wavelengths much more, while larger particles scatter all colours almost equally. Longer wavelengths scatter less, so they travel farther in the atmosphere and reach our eyes.



EXAM TIPS

- Always mention "shorter wavelength scatters more, longer wavelength scatters less."
- In diagrams, show scattering in all directions from air molecules.
- For long questions, link cause with effect (e.g., long path → blue scattered → red seen).
- Give real-life examples to score full marks.



CENTUM FOCUS

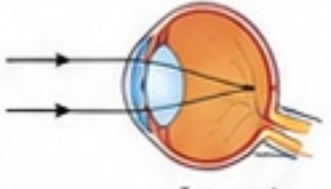
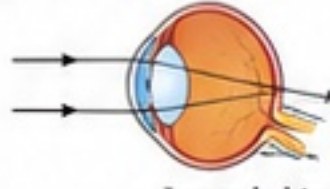
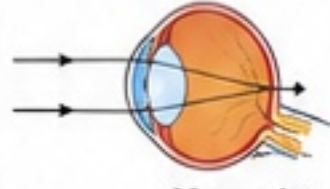
- Understand Rayleigh scattering concept deeply.
- Practice drawing path-length diagrams neatly.
- Learn the reasons for each phenomenon.
- Write structured answers with diagrams in exams.

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SECTION H - COMPARISON TABLES, CONCEPT LINKS AND APPLICATIONS

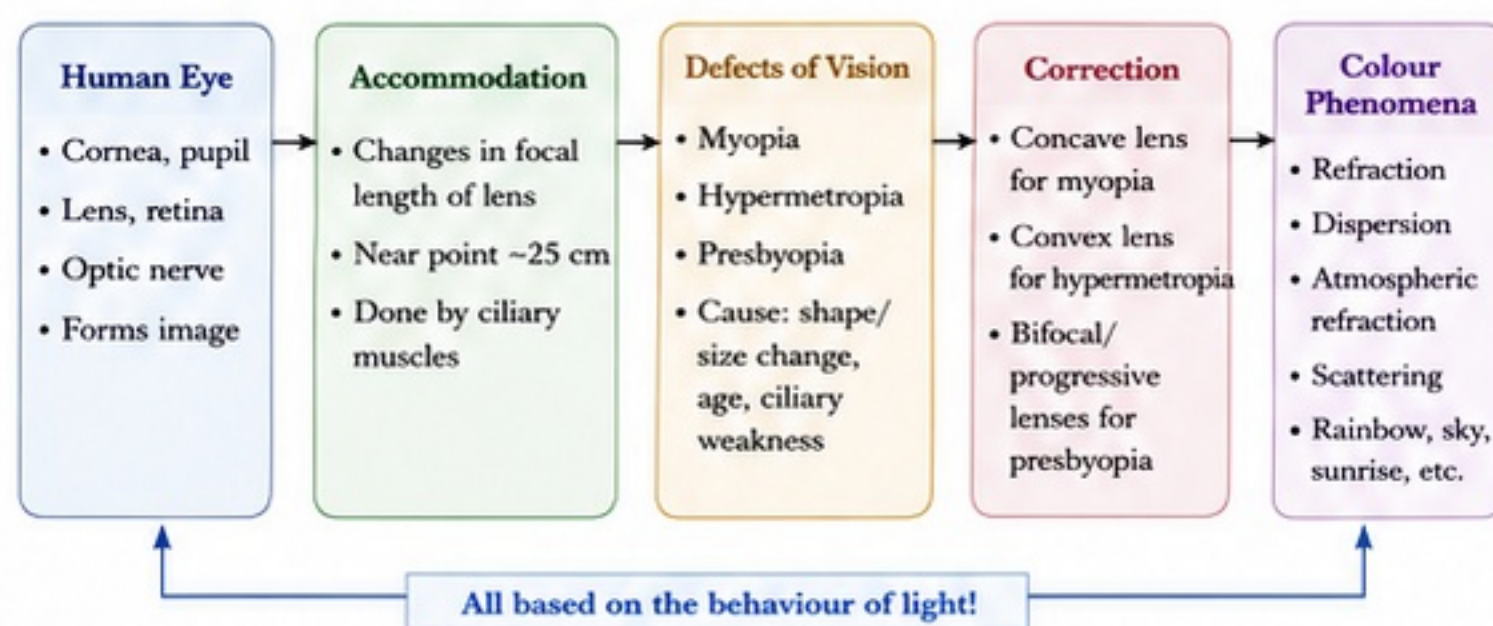
1. Comparison: Defects of Vision

Feature	Myopia (Short-sightedness)	Hypermetropia (Long-sightedness)	Presbyopia (Age-related)
Who is affected	Usually young people	Often in children or middle-aged	People above ~40 years
Cause	Eyeball too long OR eye lens too convex	Eyeball too short OR eye lens too flat	Loss of elasticity of lens; weak ciliary muscles
Far point of eye	Finite (in front of retina)	Behind retina	Shifts away progressively
Near point of eye	Normal	Farther than 25 cm	Moves farther than 25 cm
Image of distant object (without correction)	In front of retina	Behind retina	May be behind retina
Image of near object	Clear	Blurred	Blurred (especially for close objects)
Correction	Concave (diverging) lens	Convex (converging) lens	Convex lens for reading; Bifocal/Progressive lenses
Effect of correction	Shifts image back to retina	Brings image forward to retina	Helps read near objects comfortably
Simple diagram (ray idea)	 Image in front of retina	 Image behind retina	 Near point moves away

2. Comparison: Optical Phenomena in Nature

Phenomenon	What is it?	Main cause	Key feature / Example
Refraction	Bending of light when it passes from one medium to another	Change in speed of light	Straw appears bent in water; pencil looks broken in glass of water
Dispersion	Splitting of white light into its constituent colours	Different colours refract by different amounts	VIBGYOR in a rainbow or through a glass prism
Atmospheric Refraction	Bending of light by Earth's atmosphere	Gradual change in air density with height	Advanced sunrise, delayed sunset; objects appear raised
Scattering	Scattering of light by particles in the medium	Light interacts with particles much smaller than wavelength	Blue colour of sky; red colour of sunrise and sunset

3. Concept Map – How Ideas Connect



4. Daily-Life Applications / Examples

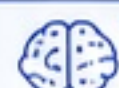
-  **Spectacles** – Correct vision by placing suitable lenses in front of the eye.
-  **Rear-view awareness** – Understand if you are short-sighted, long-sighted or presbyopic; seek timely eye check-up.
-  **Prisms in optics** – Used in periscopes, binoculars, cameras and spectrometers to deviate and analyse light.
-  **Red traffic lights look red** – Because red light is least scattered in the air and reaches our eyes.
-  **Blue sky** – Due to scattering of sunlight by tiny molecules of air.
-  **Star twinkling** – Due to atmospheric refraction and variation in air density; planets usually do not twinkle.

5. Most Important Board Points

- ✓ The human eye forms real, inverted and diminished image on the retina.
- ✓ Accommodation is the ability of the eye lens to change its focal length.
- ✓ Near point of a normal eye is about 25 cm.
- ✓ Myopia is corrected by a concave (diverging) lens.
- ✓ Hypermetropia is corrected by a convex (converging) lens.
- ✓ Presbyopia is corrected using bifocal or progressive lenses.
- ✓ Dispersion splits white light into seven colours (VIBGYOR).
- ✓ Scattering of light by air molecules is responsible for the blue colour of the sky.
- ✓ Stars twinkle due to atmospheric refraction; planets do not twinkle appreciably.
- ✓ Red colour of sunrise and sunset is due to scattering of shorter wavelengths.

6. Common Student Confusions (and Corrections)

Misconception	Correction
1. Stars and planets twinkle for the same reason.	Stars twinkle due to atmospheric refraction; planets appear as extended discs, so twinkling averages out.
2. Blue colour of sky is due to reflection of blue light.	It is due to scattering of shorter wavelengths (blue/violet) by air molecules.
3. Concave lens is used to correct long-sightedness.	Convex lens is used to correct hypermetropia (long-sightedness).
4. Convex lens is used to correct short-sightedness.	Concave lens is used to correct myopia (short-sightedness).
5. Rainbow is formed due to reflection only.	Rainbow forms due to refraction, dispersion and internal reflection in raindrops.
6. Red traffic light appears red because red is strongest colour.	Red appears red because it is least scattered and reaches us.



QUICK RECALL

- Near point of normal eye $\approx$ 25 cm.
- Concave lens → Myopia.
- Convex lens → Hypermetropia.
- Bifocal lens → Presbyopia.
- Dispersion → VIBGYOR.
- Scattering → Blue sky.



KEY IDEA

All the phenomena in this chapter are different outcomes of how light travels, bends, splits and scatters. The human eye is a precise optical instrument, and understanding light helps us correct vision and explain the beauty of nature.



EXAM TIPS

- Draw clear ray diagrams for defects and corrections.
- Label all parts of the eye neatly.
- For colour phenomena, explain the cause, not just the name.
- Use key words: refraction, dispersion, atmospheric refraction, scattering.



CENTUM FOCUS

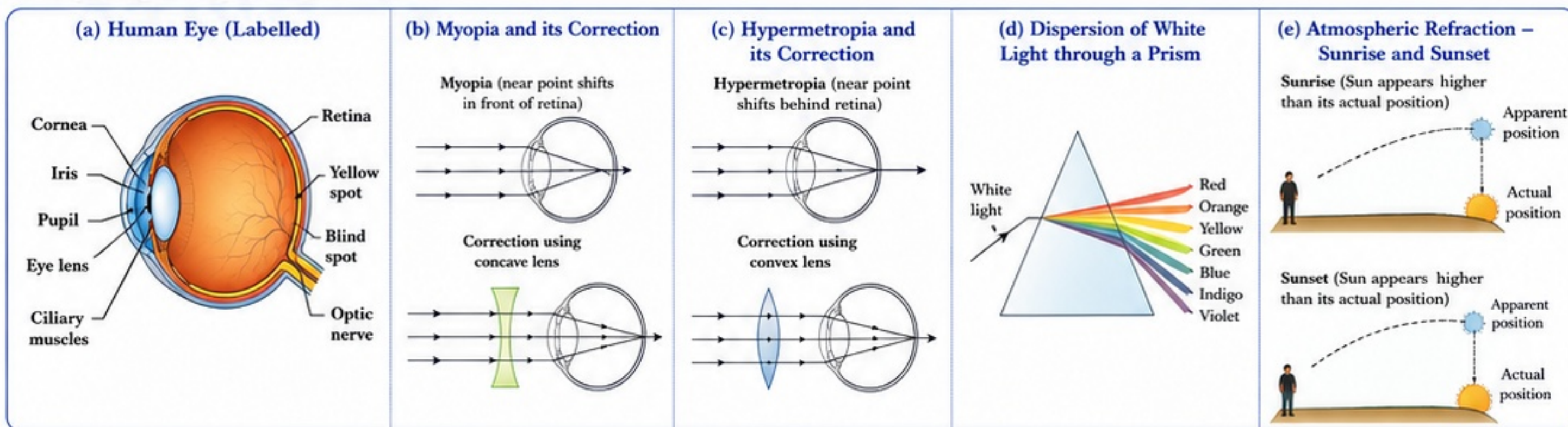
- Practise comparison tables.
- Relate every phenomenon to daily life.
- Revise diagrams until you can draw perfect ones.
- Attempt previous year board questions on this chapter.

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SECTION I - IMPORTANT DIAGRAMS, ANSWER FRAMES AND HIGH-VALUE QUESTIONS

1. Key diagrams you must know



2. Short answer frames (Board-style questions)

- **Q. What is accommodation?**
Answer Frame: Accommodation is the ability of the eye lens to change its shape so that we can see objects clearly at different distances.
- **Q. Why does the sky appear blue?**
Answer Frame: The sky appears blue due to the scattering of sunlight by the molecules of air. Blue light is scattered the most, so it reaches our eyes.
- **Q. Why do stars twinkle?**
Answer Frame: Stars twinkle due to atmospheric refraction. The continuous change in the refractive index of air layers causes the apparent position of stars to fluctuate.
- **Q. Why is the Sun reddish at sunrise and sunset?**
Answer Frame: At sunrise and sunset, sunlight travels a longer path through the atmosphere. Blue light is scattered away and mostly red light reaches our eyes.
- **Q. What is myopia and how is it corrected?**
Answer Frame: Myopia is the defect of vision in which distant objects are not seen clearly. It is corrected using a concave (diverging) lens.
- **Q. What is hypermetropia and how is it corrected?**
Answer Frame: Hypermetropia is the defect of vision in which near objects are not seen clearly. It is corrected using a convex (converging) lens.

3. Two-mark questions (with model points)

- **Q. Name the three main parts of the human eye.**
Answer: Cornea, eye lens and retina.
- **Q. What is the function of the ciliary muscles?**
Answer: They change the shape (curvature) of the eye lens for accommodation.
- **Q. What is the role of the iris?**
Answer: It controls the size of the pupil and the amount of light entering the eye.
- **Q. Define dispersion of white light.**
Answer: The splitting of white light into its constituent colours when it passes through a prism is called dispersion.
- **Q. What is the cause of twinkling of stars?**
Answer: Atmospheric refraction due to changing refractive index of air layers.
- **Q. Which colour of light is scattered the most by air molecules?**
Answer: Blue light.

4. Three-mark questions (answer skeletons)

- **Q. Explain the mechanism of accommodation in human eye.**
Answer Skeleton: (1) For near objects – ciliary muscles contract → lens becomes thicker (more convex) → focal length decreases → image on retina.
(2) For distant objects – ciliary muscles relax → lens becomes thinner (less convex) → focal length increases → image on retina.
- **Q. Explain myopia and its correction with diagram.**
Answer Skeleton: (1) Myopia – distant objects not seen clearly; image forms in front of retina. (2) Cause – eye lens too powerful/eyeball elongated. (3) Correction – use concave lens; diverges rays so the image forms on retina.
- **Q. Explain hypermetropia and its correction with diagram.**
Answer Skeleton: (1) Hypermetropia – near objects not seen clearly; image forms behind retina. (2) Cause – eye lens weak/eyeball short. (3) Correction – use convex lens; converges rays so the image forms on retina.
- **Q. Why is the Sun reddish at sunrise and sunset? Explain.**
Answer Skeleton: (1) Sunlight travels a longer distance through the atmosphere at sunrise and sunset. (2) Blue/violet light is scattered away by air molecules. (3) Red light is least scattered and reaches our eyes → Sun appears reddish.

5. Diagram-based questions – what to label

Diagram / Situation	Labels to Write
Human Eye (Labelled)	Cornea, Iris, Pupil, Eye lens, Ciliary muscles, Retina, Yellow spot, Blind spot, Optic nerve
Myopia and its Correction	Parallel rays, Image in front of retina (myopia), Concave lens, Image on retina (after correction)
Hypermetropia and its Correction	Parallel rays, Image behind retina (hypermetropia), Convex lens, Image on retina (after correction)
Dispersion through Prism	White light, Prism, Screen, Seven colours (VIBGYOR) in order
Atmospheric Refraction (Sunrise & Sunset)	Observer, Earth, Atmosphere, Actual position of Sun, Apparent position of Sun, Path of light (curved due to refraction)



QUICK RECALL

- Iris controls light entry.
- Accommodation = change in lens shape.
- Myopia → concave lens.
- Hypermetropia → convex lens.
- Blue scatters most.
- Sun appears reddish at sunrise and sunset.



KEY IDEA

The eye forms real, inverted images on the retina. Any defect due to shape/size or lens power can be corrected using suitable lenses. Light interacts with matter (scattering, refraction, dispersion) to create the colourful world we observe.



EXAM TIPS

- Draw neat, labelled diagrams.
- Start with rays → show faulty image → show correction.
- Write causes and corrections in sequence.
- Use keywords: refraction, scattering, dispersion, accommodation.



CENTUM FOCUS

- Practise all key diagrams daily.
- Write short answers in crisp bullet points.
- Revise common Board questions from past papers.
- Attempt diagram-based questions for full marks.

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SECTION J - MASTER REVISION SHEET, NCERT TRAPS AND CENTUM FOCUS

1. COMPACT DEFINITIONS

Term	Definition
Accommodation	Ability of the eye lens to change its curvature to focus near or far objects clearly on the retina.
Retina	Light-sensitive layer at the back of the eye where the image is formed.
Pupil	Opening in the centre of the iris that allows light to enter the eye.
Cataract	Clouding of the eye lens, causing blurred or dim vision.
Myopia	Defect in which distant objects are not seen clearly; image forms in front of the retina.
Hypermetropia	Defect in which near objects are not seen clearly; image forms behind the retina.
Presbyopia	Age-related loss of accommodation; difficulty in seeing nearby objects.
Dispersion	Splitting of white light into its constituent colours by a prism due to refraction.
Spectrum	Band of seven colours (VIBGYOR) produced by dispersion of white light.
Atmospheric refraction	Bending of light as it passes through layers of the atmosphere with different densities.
Scattering	Deflection of light in different directions by very small particles of the medium.
Tyndall effect	Scattering of light by colloidal particles, making the path of light visible.

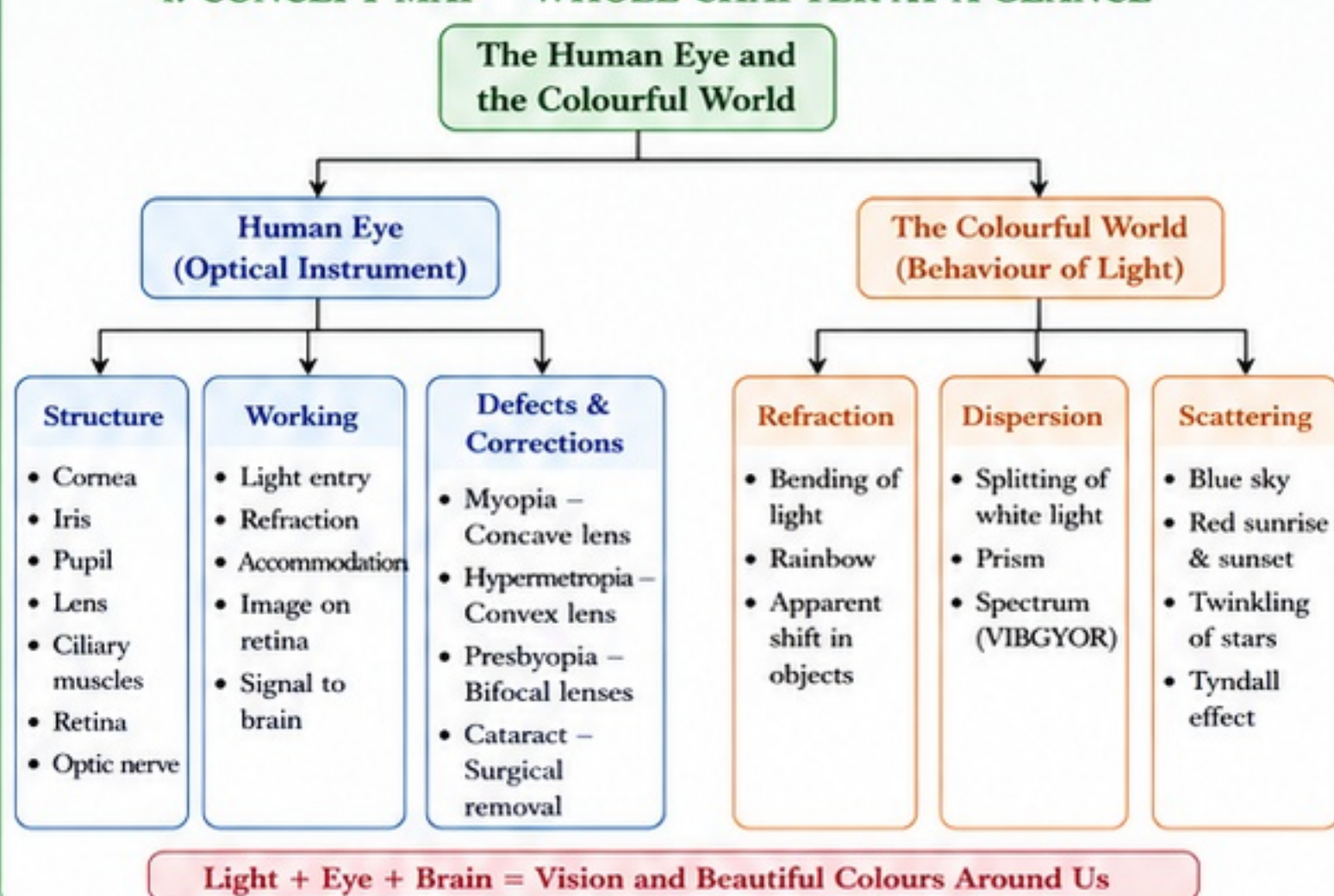
2. MASTER SUMMARY – THE WHOLE CHAPTER IN ONE VIEW

- **Structure of the Eye:** Cornea (protection & refraction), aqueous humour (nourishment), iris (controls light), pupil (light entry), eye lens (focus), ciliary muscles (change lens shape), vitreous humour (maintains shape), retina (image formation), optic nerve (transmits signals).
- **Working of the Eye:** Light enters → cornea → pupil → lens → vitreous humour → retina. Ciliary muscles change lens curvature (accommodation). Image formed on retina; optic nerve sends signals to the brain.
- **Defects of Vision:** Myopia (near-sightedness) – corrected by concave lens. Hypermetropia (far-sightedness) – corrected by convex lens. Presbyopia (age-related) – corrected by bifocal/progressive lenses. Cataract – treated by removal of cloudy lens and insertion of artificial lens.
- **Power of Accommodation:** Range of distance between the nearest and farthest points clearly visible by the eye (~25 cm to ∞ for normal eye).
- **Refraction of Light:** Bending of light on passing from one medium to another due to change in speed.
- **Dispersion of Light:** Splitting of white light into seven colours by a prism due to different refractions for different wavelengths.
- **Atmospheric Refraction:** Causes twinkling of stars, advanced sunrise and delayed sunset.
- **Scattering of Light:** Responsible for blue colour of sky and red colour of sunrise/sunset.
- **Tyndall Effect:** Path of light becomes visible in colloidal solutions, smoke, fog etc. (e.g., torch light in dark room).
- The Colourful World is explained by the behaviour of light – refraction, dispersion, scattering and human eye's ability to sense colours and form images.

3. NCERT TRAPS AND MISCONCEPTIONS

Trap / Misconception	Correct Explanation
Pupil increases in bright light.	False. Pupil constricts (becomes smaller) in bright light and dilates in dim light.
Image formed on the cornea.	False. Cornea only refracts light; image is formed on the retina.
We see objects because light from them travels in straight lines only.	Partly true. Light travels in straight lines in a homogeneous medium, but refraction changes direction.
Myopia is corrected using a convex lens.	False. Myopia is corrected using a concave lens.
Hypermetropia is corrected using a concave lens.	False. Hypermetropia is corrected using a convex lens.
Cataract is due to damage of the retina.	False. Cataract is due to clouding of the lens.
Scattering makes the sky red.	False. Scattering makes the sky blue; red colour at sunrise/sunset is due to longer path length and scattering of blue light.
Tyndall effect occurs in true solutions.	False. It occurs in colloids, not in true solutions.
White light is single-coloured.	False. White light is a mixture of seven colours.
Optic nerve produces the image.	False. Optic nerve only transmits signals; retina forms the image.

4. CONCEPT MAP – WHOLE CHAPTER AT A GLANCE



5. 10 MUST REMEMBER

1. The image is finally formed on the retina.
2. Ciliary muscles change the curvature of the lens (accommodation).
3. Myopia → concave lens; Hypermetropia → convex lens.
4. Presbyopia occurs due to weak ciliary muscles with age.
5. Cataract is clouding of the lens; treated by surgery.
6. Dispersion of white light by prism → spectrum (VIBGYOR).
7. Atmospheric refraction → advanced sunrise, delayed sunset, twinkling of stars.
8. Scattering of light → blue sky, red sunrise/sunset.
9. Tyndall effect makes the path of light visible in colloids.
10. Light is essential for vision; without light we cannot see anything.

6. FINAL CENTUM CHECKLIST ✓

- ☐ I can label and explain all parts of the human eye.
- ☐ I can draw the diagram of the eye and its working.
- ☐ I can explain accommodation with diagram.
- ☐ I can state defects of vision and their corrections.
- ☐ I can define and give examples of refraction.
- ☐ I can explain dispersion and draw the spectrum.
- ☐ I can explain atmospheric refraction with examples.
- ☐ I can explain scattering and its effects.
- ☐ I can explain Tyndall effect with examples.
- ☐ I can solve numerical problems on refraction.

7. BEFORE EXAM – QUICK GLANCE

- 👁️ Know the eye parts and their functions.
- 🔍 Understand accommodation clearly.
- 👓 Remember defects and corrections.
- 🌟 Be ready to draw neat diagrams.
- 🌈 Relate light phenomena with daily life examples.
- ☁️ Practice important diagrams and labelled figures once more.
- 📚 Revise NCERT in-text and end questions.



Revise smart, draw diagrams neatly, and score full marks.

